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DATA VISUALIZATION TECHNIQUES

UNIT - IV

Pulla Manoj Kumar
M.Tech (CSE), B.Tech (CSE)

Assistant Professor
Department of CSE (Data Science)
ACE Engineering College, Hyderabad

Mosaic Plots and Their Variants

- **Definition and Construction**
- **Interpreting Mosaic Plots**
  - Probabilities in Mosaic Plots
  - Visualizing Interaction Effects
- **Variants**
  - Doubledecker Plots
  - Fluctuation Diagrams
  - Others
- **Related Work and Generalization**
  - Treemaps
  - Trellis Plots
  - Pivot Tables
- **Implementations**

Introduction to Mosaic Plots

- **Historical Origin:** Mosaic plots were formally introduced by Hartigan and Kleiner in 1981.
- **Purpose of the Technique:** They were developed as a specialized graphical method for the visualization of multi-way contingency tables.
- **Reason for the Name:** The term “mosaic plot” comes from its visual resemblance to traditional mosaic artwork.
- **Graphical Structure:** The plot consists of groups of interconnected rectangles, where each rectangle corresponds to a specific cell within a contingency table.
- **Interpretation Complexity:** Both the physical size and the geometric position of the rectangles carry statistical meaning, making mosaic plots one of the more advanced graphical tools used in statistical analysis.

The Core Advantages of Mosaic Visualization

- **Preservation of Categorical Information:** A major advantage of mosaic plots is their ability to preserve all categorical information contained within multivariate contingency tables.
- **Detailed Representation:** Every rectangle in the mosaic corresponds directly to a cell in the contingency table, ensuring that the complete data structure is represented.
- **Simultaneous Overview:** While retaining detailed information, mosaic plots also provide a high-level overview of relationships between categorical variables.
- **Identification of Patterns:** This visualization allows researchers to quickly detect broader trends and associations between categories.
- **Dimensional Flexibility:** Although most mosaic plots are displayed in two or three dimensions due to space limitations, the theoretical framework allows representation of many categorical dimensions.

Definition and Construction Process

- **Categorical Variables:** Consider a set of categorical variables $X_1, X_2, \ldots, X_p$.
- **Number of Categories:** For each variable X_i, let c_i denote the total number of categories.
- **Initial Rectangle:** The mosaic construction begins with a single rectangle r_0 having width w_0 and height h_0.
- **Hierarchical Splitting:** This rectangle is divided into c_i parts according to the category proportions of variable X_i.
- **Iterative Subdivision:** The process continues iteratively by splitting either the width or height of the previous rectangles into proportional rectangles r_{ij} where $j = 1, 2, \ldots, c_i$.
- **Maintaining Proportions:** At every step, the alternate dimension remains constant so that the relative proportions of the categories are preserved.

Construction of a p-dimensional Mosaic Plot

		Class				
		1st	2nd	3rd	Crew	Totals
Sex	f	145	106	196	23	470
	m	180	179	510	862	1731
Totals		325	285	706	885	2201

Figure 1: Breakdown of Gender versus Class for those aboard the Titanic (Dawson, 1995)

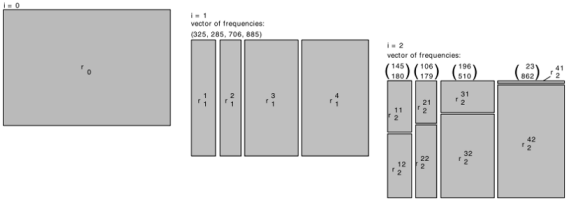


Figure 2: Stepwise construction of a two-dimensional mosaicplot of Gender distribution across different classes based on the data given in Table 1

Construction of a p-dimensional Mosaic Plot

Class:	First		Second		Third		Crew	
Gender:	Female	Male	Female	Male	Female	Male	Female	Male
Survived:								
yes	141	62	93	25	90	88	20	192
no	4	118	13	154	106	422	3	670
Survival Rate (in %)	97	34	88	14	46	17	87	22

Figure 1: Titanic survivors by class and gender, showing large survival differences between men and women

Gender:	Female				Male			
Class:	First	Second	Third	Crew	First	Second	Third	Crew
Survived:								
yes	141	93	90	20	62	25	88	192
no	4	13	106	3	118	154	422	670
Survival Rate (in %)	97	88	46	87	34	14	17	22

Figure 2: Titanic survivors by gender and class; survival decreases with lower passenger class

Construction of a p-dimensional Mosaic Plot

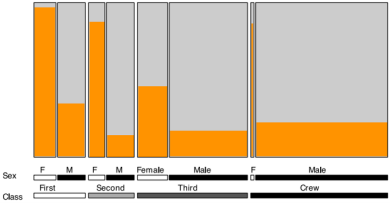


Figure 1: 3D mosaic plot of Titanic data showing higher survival rates for women across all classes

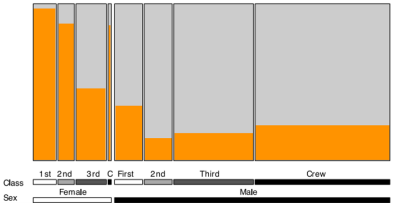


Figure 2: 3D mosaic plot of Titanic data showing higher survival for women and extremely low survival for second-class men.

The Critical Role of Variable Order

- **Hierarchical Construction:** The graphical construction of a mosaic plot follows a strict hierarchical sequence.
- **Relationship to Contingency Tables:** This hierarchy mirrors the structure of the corresponding multiway contingency table.
- **Effect of Changing Order:** Changing the order in which variables are introduced significantly alters the visual structure of the mosaic plot.
- **Different Analytical Insights:** Different orderings emphasize different statistical relationships within the same dataset.
- **Methodological Challenge:** Selecting an appropriate ordering of variables is often one of the most difficult aspects for analysts when constructing mosaic plots.

Understanding Plot Structure and Frame

- **Need for Structural Understanding:** Multiple mosaic plots may display the same numerical data but represent different contingency table structures.
- **Role of the Frame:** To interpret the plot correctly, analysts must examine the structural frame of the mosaic plot.
- **Meta-information Representation:** The frame captures important information about the variables involved in the visualization.
- **Variable Characteristics:** It specifies the possible values each variable can take and the order in which these categories appear.
- **Directional Splitting:** The structure also determines whether variables are introduced through horizontal or vertical splits in the plot.

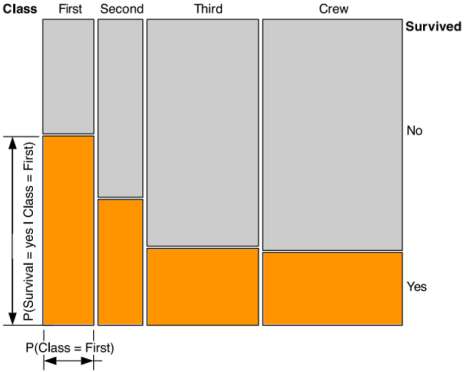
Interpreting Mosaic Plots: Probabilities

- **Area-Based Representation:** Mosaic plots represent contingency tables using rectangular tiles whose areas correspond to cell sizes.
- **Tile Interpretation:** Each rectangle represents a category combination and its size reflects the relative frequency.
- **Proportional Representation:** Tile area is proportional to the observed counts in the contingency table.
- **Joint Probability:** The area of a tile represents the joint probability of the corresponding categories.

$$\text{Area} \propto \text{height} \times \text{width}$$

$$P(\text{Survival} = \text{yes} \cap \text{Class} = \text{1st})$$

Interpreting Mosaic Plots: Probabilities



2D mosaic plot of Class vs. Survival. Tile width shows class probability; height shows conditional probability of (non-)survival.

Directional Splits in Mosaic Plots

- Variables divide the plot through **vertical** or **horizontal** splits.
- **Vertical split set:** $v(i)$ denotes variables among the first i variables producing vertical divisions.
- **Horizontal split set:** $h(i)$ denotes variables producing horizontal divisions.

$$\emptyset = v(0) \subset v(1) \subset \cdots \subset v(p)$$

$$v(p) \cup h(p) = \{X_1, X_2, \ldots, X_p\}$$

- In the Hartigan–Kleiner method: even-index variables → vertical splits, odd-index variables → horizontal splits.

Widths and Heights as Probabilities

- After the ith split, rectangle dimensions represent conditional probabilities.
- **Width**

$$\text{Width} = P(h(i) \mid v(i))$$

- **Height**

$$\text{Height} = P(v(i) \mid h(i))$$

- For a horizontal split:

$$P(h(i)|v(i)) = P(X_i|v(i)) \times P(h(i-1)|v(i))$$

- Rectangle dimensions therefore visualize conditional relationships.

Visualizing Interaction Effects

- Association between categorical variables can be measured using the **odds ratio**.
- Consider two binary variables forming a 2×2 table.

$$\pi_{ij} = P(X = X_i \cap Y = Y_j)$$

- **Odds Ratio**

$$\theta = \frac{\pi_{12}\pi_{21}}{\pi_{11}\pi_{22}}$$

  - $\theta \approx 0 \rightarrow$ strong negative association
  - $\theta = 1 \rightarrow$ independence

Geometric Representation of Interactions

- Interaction can be visually detected through differences in conditional probabilities.
- **Geometric Difference**

$$d = \frac{m_{12}}{m_{11} + m_{12}} - \frac{m_{22}}{m_{21} + m_{22}}$$

- Approximate relation with odds ratio:

$$d \approx -0.25 \log \theta$$

- Geometric patterns in mosaic plots indicate the strength of interaction.

Connection to Loglinear Models

- Mosaic plots help visually identify suitable **loglinear models**.
- They also highlight deviations and residual patterns.
- **Independence Model**

$$\log m_{ij} = \mu + \lambda_i^X + \lambda_j^Y$$

  - μ : grand mean
  - λ : variable effects
- Interaction parameter:

$$\lambda_{22}^{XY} = \log \theta$$

Advanced Interactions and High-Dimensionality

- **Three-Way Statistical Interactions:** A three-way interaction among variables (X, Y, Z) exists when the two-way interaction between X and Y changes across the different levels of Z.
- **Conditional Interaction Concept:** This means the association between two variables is not constant but depends on the category of the third variable.
- **Conditional Odds Ratio:** For a fixed level k of Z, the conditional odds ratio is calculated as

$$\theta_{ij(k)} = \frac{m_{i+1,j,k}\, m_{i,j+1,k}}{m_{i,j,k}\, m_{i+1,j+1,k}}$$

- **Comparing Conditional Odds Ratios:** To evaluate interaction across levels, the logarithms of the conditional odds ratios are compared.
- **Three-Way Interaction Formula:**

$$\log \theta_1 - \log \theta_2 = \log(\theta_1/\theta_2) = \log \theta_{XYZ}$$

- **Interpretation:** If $\theta_{XYZ} = 1$, the conditional odds ratios are equal and there is no three-way interaction.

Multivariate Extensions of Odds Ratios

- **Multiple Categories:** For variables with more than two categories, interactions are studied using generalized odds ratios.
- **Adjacent Cell Quadruples:** Interaction is analyzed using sets of four neighboring cells in contingency tables.
- **Generalized Odds Ratio**

$$\theta_{ij} = \frac{m_{i+1,j}\, m_{i,j+1}}{m_{ij}\, m_{i+1,j+1}}$$

- **Geometric Interpretation:** Differences in vertical heights of rectangles in mosaic plots indicate interaction effects.
- **Approximation**

$$-4d_i \approx \log \theta_i$$

$$\log \theta_i = \log \left(\frac{m_{(i+1)1} m_{i2}}{m_{i1} m_{(i+1)2}} \right)$$

Introduction to Mosaic Variants

- **Dependence on Construction Process:** In standard mosaic plots, the shapes, sizes, and spatial arrangement of tiles are determined by the hierarchical splitting algorithm.
- **Purpose of Structural Variations:** Researchers often modify these graphical structures to explore different statistical perspectives within the same dataset.
- **Exploratory Visualization Strategy:** Changing the graphical framework can highlight particular statistical relationships more clearly.
- **Simplifying Visual Interpretation:** These modifications may also reduce visual complexity and make interpretation easier.
- **Two Major Categories of Variants:**
  - Structural variants (e.g., Doubledecker plots)
  - Layout or aesthetic variants (e.g., Fluctuation diagrams)

Variants: Doubledecker Plots

- **Structural Modification:** Doubledecker plots represent a structural variation of mosaic plots.
- **Different Splitting Strategy:** Instead of alternating vertical and horizontal splits, all base categories are split horizontally.
- **Mathematical Definition:**

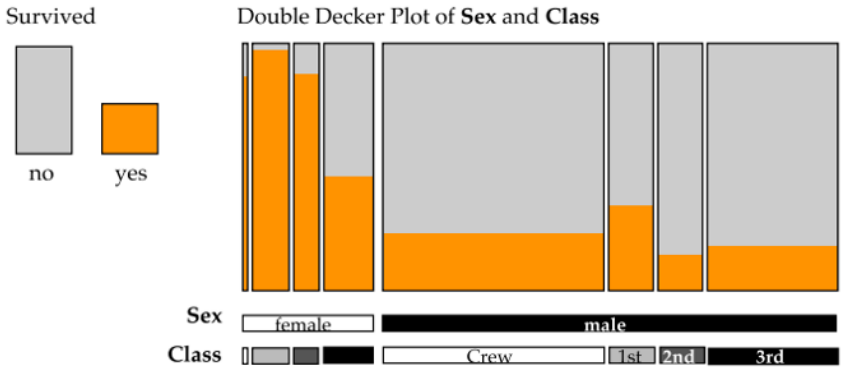
$$v(i) = \emptyset, \quad i = 1, \dots, p-1$$
$$v(p) = \{X_p\}$$

- **Uniform Baseline Layout:** All baseline bins appear with identical vertical height and are placed side-by-side.
- **Focus on Conditional Probability:**

$$P(X_p \mid X_1, \dots, X_{p-1})$$

- **Interpretation Advantage:** This arrangement clearly emphasizes conditional probability distributions.

Doubledecker Plots



Doubledecker plot of the Titanic data. Survivors are highlighted. he highest survival rates appear for the women in the crew and in the first and second classes

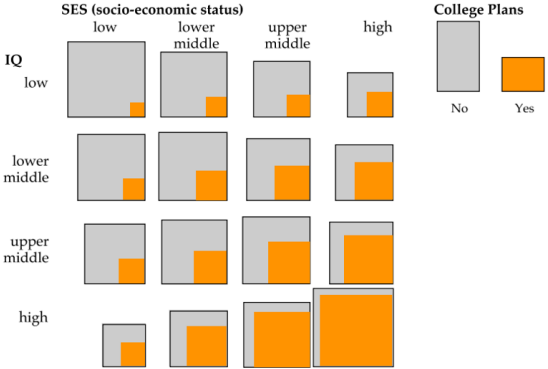
Enhancing Doubledecker Usability

- **Improved Labeling:** One major advantage of doubledecker plots is the ease of labeling categorical combinations.
- **Striped Bar Representation:** A striped bar is placed beneath the plot to represent the categories of each variable.
- **Alternating Category Colors:** Different shades are used within these bars to distinguish category values.
- **Reading Category Combinations:** To identify the variable combination for a bin, one simply traces a vertical line downward through the labeled bars.
- **Practical Benefit:** This method greatly simplifies interpretation compared with traditional mosaic plots.

Variants: Fluctuation Diagrams

- **Alternative Visualization Framework:** Fluctuation diagrams modify the spatial layout of mosaic plots.
- **Area Representation:** The area of each rectangle still represents the number of cases in a category.
- **Fixed Grid Structure:** Instead of filling space freely, rectangles are positioned on a uniform rectangular grid.
- **Fixed Position of Cells:** The bottom-right corner of each bin is fixed to a predetermined grid location.
- **Size Adjustment:** Both the height and width of each rectangle shrink proportionally to represent the cell size.
- **Square-Root Scaling:** For square grids, the dimensions of the rectangle become proportional to the square root of the cell frequency.

Fluctuation Diagrams



Two-dimensional mosaicplot in fluctuation mode. Highlighting shows pupils who intend to go to college. hree trends are recognizable visually

Applications of Fluctuation Diagrams

- **Emphasizing Large Data Values:** Fluctuation diagrams visually emphasize heavily populated bins by shrinking the representation of smaller values.
- **Visual Impact of Sparse Cells:** Bins with very small counts shrink considerably, while completely empty bins may visually disappear.
- **Matrix Visualization Applications:** This property makes fluctuation diagrams highly useful for visualizing large matrices.
- **Example: Correlation Matrices:** They are often applied to large 36×36 correlation matrices to reveal seasonal or structural correlation patterns.
- **Standardizing Case-by-Variable Matrices:** When analyzing Bertin plots with percentage values, analysts frequently standardize the data.
- **Standardization Formula:**

$$X \leftarrow \frac{X}{\sigma_X}$$

This transformation helps reveal subtle structural patterns across observations.

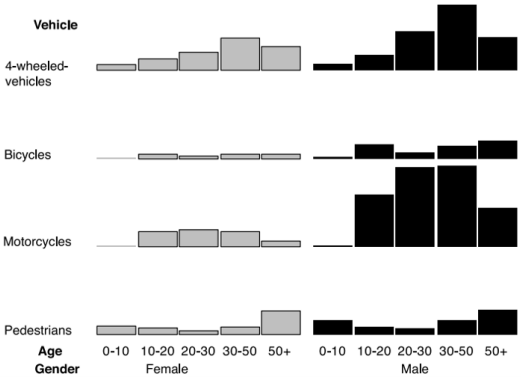
Reordering in Fluctuation Diagrams

- **Validating Clustering Results:** Fluctuation diagrams can visually validate the results of hierarchical clustering algorithms such as Complete Linkage.
- **Grouping Similar Objects:** Rows and columns can be reordered based on clustering outputs so that similar objects appear close together.
- **Spatial Grouping Effect:** This spatial grouping helps reveal hidden hierarchical relationships within the dataset.
- **Purity Function Concept:** Optimal ordering corresponds to minimizing a mathematical objective known as the purity function.
- **Interactive Data Exploration:** Modern visualization software allows dynamic reordering of rows and columns based on computed scores.

Other Variants: Multiple Bars

- **Relationship to Fluctuation Diagrams:** The multiple bar display represents a graphical variant closely related to fluctuation diagrams.
- **Unidirectional Shrinking:** Instead of shrinking the area of rectangles proportionally, the tiles shrink along only one dimension.
- **Barchart-Like Appearance:** This modification causes the resulting visualization to resemble a multi-level bar chart.
- **Hierarchical Dependence:** The hierarchical structure of variables determines the arrangement of the bars.
- **Effect of Variable Order:** Changing the order of variables shifts the positions of the smaller embedded bar charts without altering their sizes.

Multiple Bars



Multiple-bar display of Accident Victims data (Bertin, 1967). More men than women are victims, especially in motorcycle accidents, while for women, four-wheeled vehicles are the main source. Exceptions are girls under 10 and women over 50, where most accidents involve pedestrians.

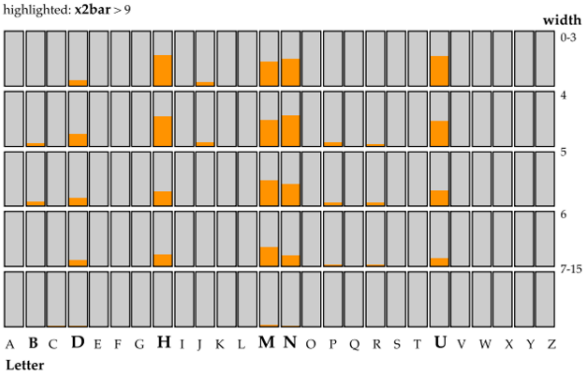
Other Variants: Same-Bin-Size Displays

- **Uniform Bin Size Concept:** In the same-bin-size display, every bin is forced to have the same visual size.
- **Weight Adjustment Strategy:** This is achieved by assigning observation weights inversely proportional to the number of cases in each cell.
- **Weighting Formula:**

$$W_i \propto \frac{1}{\text{number of cases in the cell}}$$

- **Comparing Conditional Distributions:** This approach simplifies the comparison of conditional distributions across categories.
- **Standardizing Vertical Scales:** Uniform bin sizes eliminate the difficulty of comparing distributions with different vertical scales.
- **Diagnostic Capability:** The method is particularly useful for identifying empty-bin patterns in sparse multidimensional datasets.

Same-Bin-Size Displays

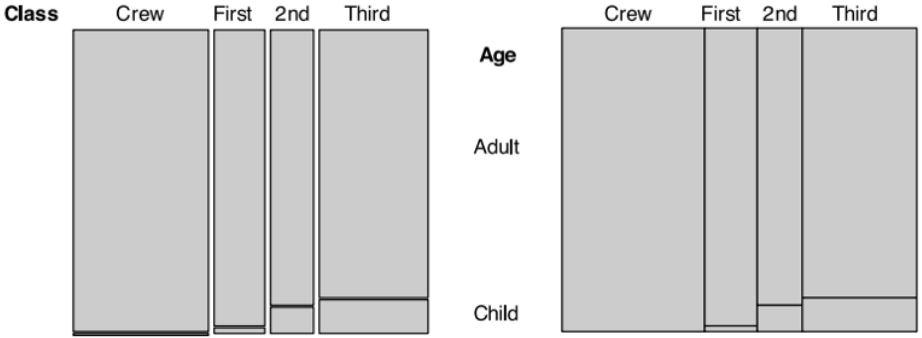


Two-dimensional mosaicplot of the Letter Recognition data. Due to the same-bin-size representation, the pattern for the highlighted cases is clearly visible

Other Variants: Mondrian Displays

- **Space-Filling Nature of Mosaic Plots:** Traditional mosaic plots separate rectangles with thin spaces for visual clarity.
- **Mondrian Display Concept:** When these separating spaces are removed completely, the resulting visualization is called a Mondrian display.
- **Compact Visualization:** This approach creates a highly compact graphical representation.
- **Efficient Use of Screen Space:** Mondrian displays are particularly useful when working with datasets containing many categories.
- **Limitation of the Method:** Empty category combinations disappear entirely because no visual gaps are preserved.

Mondrian Displays

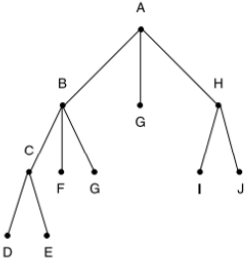


Regular (let) and Mondrian (right) mosaicplots of the Titanic data. Class is plotted versus Age. he empty crew/child cell disappears from the Mondrian display

Related Work: Treemaps

- **Historical Development:** Treemaps were introduced by Johnson and Shneiderman in 1991.
- **Original Application:** They were developed to visualize large hierarchical file structures stored on computer hard disks.
- **Node Representation:** Each node in the hierarchy is displayed as a rectangle.
- **Size Representation:** The size of each rectangle is proportional to the numerical value or disk space represented by the node.
- **Hierarchical Visualization Strength:** Treemaps can represent very complex hierarchical structures.
- **Comparison with Mosaic Plots:** While contingency tables can always be mapped to tree structures, arbitrary hierarchical trees cannot always be reduced to simple contingency tables.

Treemaps



Tree (let) and accompanying treemap (right). In this example all nodes have equal weight. Only leaves of the tree are visible as tiles; the other nodes determine the structure

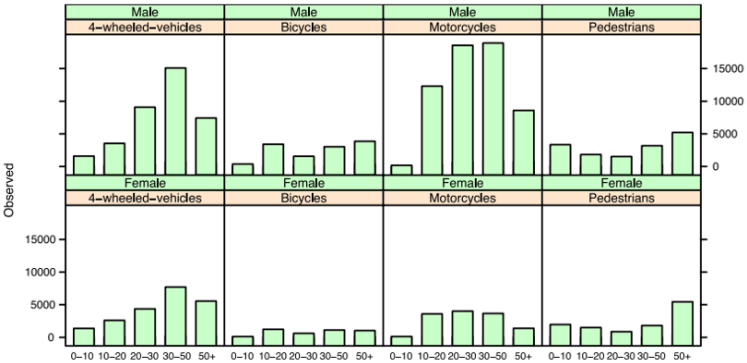
Related Work: Trellis Plots

- **Panel-Based Visualization:** Trellis plots follow a concept related to mosaic plots but present the data through a panel-based layout.
- **Conditioning Through Panels:** Data subsets are separated and displayed within individual panels arranged systematically across rows and columns.
- **Flexible Graphical Representation:** Unlike mosaic plots that are restricted to rectangular area displays, each panel in a trellis layout can display different graphical techniques.
- **Multiple Plot Types:** Panels may contain scatterplots, boxplots, dot plots, or other statistical graphics depending on the analytical objective.
- **Mathematical Formula Representation:** Trellis displays are commonly expressed using formula notation.
- **Trellis Formula Example:**

$$y \sim x \mid g_1 * g_2 * \cdots * g_p$$

This formula corresponds conceptually to an inverted mosaic structure ordered as

Trellis Plots



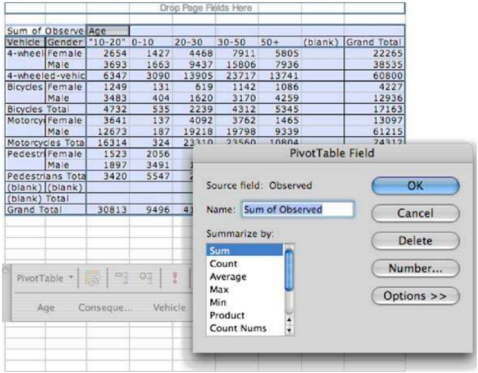
Trellis display of the Accident Victims data.

Related Work: Pivot Tables

- **Software Popularization:** Pivot tables are widely used numerical analysis tools available in spreadsheet software such as Microsoft Excel.
- **Relationship to Mosaic Plots:** Although primarily numerical rather than graphical, pivot tables share a structural similarity with mosaic plots.
- **Handling Meta-Variables:** Pivot tables store the ordering of row and column variables separately in structured lists.
- **Reduced Meta-Information:** Compared with mosaic plots, pivot tables capture slightly less structural meta-information about variable hierarchy.
- **Marginal Data Availability:** A major advantage of pivot tables is the easy computation of marginal totals.
- **Examples of Marginal Statistics:**
  - Row totals
  - Column totals
  - Grand totals

These summaries are visually implicit in mosaic plots but explicitly available in pivot tables.

Pivot Tables



Drop Page Fields Here								
Sum of Observe	Age							
Vehicle	Gender	"10-20"	0-10	20-30	30-50	50+	(blank)	Grand Total
4-wheel	Female	2654	1427	4468	7911	5805		22265
	Male	3693	1663	9437	15806	7936		38535
4-wheeled-vehic		6347	3090	13905	23717	13741		60800
Bicycles	Female	1249	131	619	1142	1086		4227
	Male	3483	404	1620	3170	4259		12936
Bicycles Total		4732	535	2239	4312	5345		17163
Motorcy	Female	3641	137	4092	3762	1465		13097
	Male	12673	187	19218	19798	9339		61215
Motorcycles Tota		16314	324	23310	23560	10804		74312
Pedestr	Female	1523	2056					
	Male	1897	3491					
Pedestrians Tota		3420	5547					
(blank)	(blank)							
(blank) Total								
Grand Total		30813	9496	4				

Pivot table for the Accident Victims data. Front: dialog window showing the summarizing options available for in Excel's pivot tables

Software Implementations

- **R Statistical Environment:** Mosaic plots and their extensions are strongly supported within the R programming environment.
- **Key Implementations:** Important implementations and visualization functions were developed by researchers such as John Emerson.
- **Interactive Visualization Tools:** Advanced visualization environments allow dynamic interaction with mosaic plots.
- **Examples of Interactive Systems:**
  - MANET
  - Mondrian
  - KLIMT
- **Dynamic Data Exploration:** These systems allow variables to be reordered interactively and visualizations to be recomputed in real time.
- **Evolution of Graphical Statistics:** Modern developments in graphical statistics combine ideas from traditional mosaics and newer visualization techniques to enhance analytical power.